EVIDENCE LEVEL: E0 STATUS: CONCEPTUAL_PREVIEW MODULE: 4

The Cost of Commitment

Commitment is not a uniform act. This article distinguishes types of

This derived research essay is authored under **Editorial Derivation Brief

1. The Spectrum of Commitment

In market formation systems, commitment is the bridge that transforms private

When platform architects fail to distinguish between exploratory interest,

To establish fair demand formation, DC2SC Module 04 introduces a rigorous

2. Ratification Is Not Commitment

The core constitutional axiom formulated in Module 04 states: **Ratification is

Ratification (Stage 02): An explicit, contestable signal by a participant

Conditional Commitment (Stage 03): A provisional escrow of resources or

Binding Execution (Stage 04): The final contractual close where exchange

Conflating Stage 02 with Stage 04 is the primary failure mode of coercive group

3. The Burden of Dissolution

The central test of any coordination mechanism is not how it behaves during

Collective market positions frequently dissolve due to:

Insufficient cohort density to meet supplier minimums.

2. Rejection or opportunistic counter-offers by suppliers.

3. Temporal drift that renders the original constraint windows obsolete.

In an ethical architecture governed by **Continuous Rights Gates**, position

during prolonged negotiation windows.

SCOLIA operates strictly as a bounded stress-test environment, not a commercial booking service.

4. Failure Conditions Tracked

Under the **DC2SC Failure-First Framework**, Module 04 continuously monitors two

Premature Lock-in (FAIL-04): Participants are penalized for withdrawing from

Asymmetric Dissolution Loss (FAIL-11): Intermediaries or suppliers retain

5. Research Handoff to Module 05

Protecting participants from asymmetric commitment costs requires formalizing

It does not constitute commercial claims, supplier commitments, or live transaction guarantees.

MATHEMATICAL INVARIANT // INVARIANT 4.1: THE ESCROW COMMITMENT DISSOLUTION ASYMMETRY (Eq. 4.

$$Loss_{buyer}(tau) = INT_0^{tau} (R_{out}(t) - R_{escrow}(t)) dt + lambda * 1_{{breach}} * E_{sunk}$$

Proves that unconditioned commitment creates monotonic downside decay for participants

[FORMALLY VERIFIED MATHEMATICAL SPECIFICATION PEER CITATION READY]

CONSTITUTIONAL RESEARCH BOUNDARY & NON-CLAIM DISCLOSURE

This publication presents conceptual research and formal invariants at Evidence Level E0.